



Darshan University

A Project Report on

“Loan Management System”

Under the subject

Software Engineering (2301CS405)

B. Tech, Semester – IV

Computer Science & Engineering Department

Submitted By

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Internal Guide

Dean-DIET

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DECLARATION

We hereby declare that the SRS, submitted along with the **Software Engineering (2301CS405)** for entitled “**Loan Management System**” submitted in partial fulfilment for the Semester-5 of **Bachelor Technology (B. Tech)** in **Computer Science and Engineering (CSE)** Department to Darshan University, Rajkot, is a record of the work carried out at **Darshan University, Rajkot** under the supervision of R. B. Gondaliya and that no part of any of report has been directly copied from any students’ reports, without providing due reference.

(Kalp Vipul Viradia)

Student’s Signature

Date: 01/03/2025



**Computer Science & Engineering
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CERTIFICATE

This is to certify that the SRS on “**Loan Management System**” has been satisfactorily prepared by **Kalp Vipul Viradia (23010101299)** under my guidance in the fulfillment of the course **Software Engineering (2301CS405)** work during the academic year 2024-2025.

Internal Guide

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Thus, in conclusion to the above said, I once again thank the faculties and members of **Darshan University** for their valuable support in completion of the project.

Thanking You

Kalp Vipul Viradia

ABSTRACT

The Loan Management System (LMS) is a digital solution designed to automate and streamline the entire loan process for financial institutions. This system simplifies loan application, approval, repayment, and reporting, reducing manual efforts and ensuring accuracy. It offers a user-friendly interface for both clients and loan officers, enabling clients to apply for loans securely and track their repayment schedules. Once a loan application is submitted, loan officers can review, approve, or reject it, with the system automatically updating the status. Clients can view their outstanding balances and make payments through integrated payment gateways.

Administrators can generate comprehensive reports on loan disbursements, repayments, and overdue loans, helping in effective resource management. The system's secure login and data privacy features ensure that client information is protected. By reducing paperwork, minimizing human error, and improving operational efficiency, the LMS enhances both the loan management process and the customer experience, providing a seamless and efficient solution for loan handling.

Main purpose of this system is to reduce human efforts as much as possible.

Table of Contents

List of Figures.....	I
List of Tables.....	II
1 Introduction	1
1.1 Product perspective	1
1.2 Product features	1
1.2.1 There are three different users who will be using this product:	1
1.2.2 The features that are required for the Admin are:.....	1
1.2.3 The features that are required for the Loan officer are:	1
1.2.4 The features that are required for the Customer are:.....	1
1.3 Functional Requirement	2
1.3.1 Admin.....	2
1.3.2 Loan Officer.....	2
1.3.3 Customer.....	2
1.4 Non-Functional Requirement	3
1.4.1 Usability:	3
1.4.2 Accuracy:.....	3
1.4.3 Availability:.....	3
1.4.4 Maintainability:.....	3
2 Design and Implementation Constraints	4
2.1 Use case diagram	4
2.2 Activity diagram and Swimlane diagram	4
2.3 Sequence diagram.....	7
2.4 State diagram.....	9

2.5	Class diagram	9
2.6	Data flow diagram.....	11
2.6.1	Context diagram (level-0)	11
2.6.2	DFD Level-1	12
2.6.3	DFD Level-2	12
3	External interface requirement (Screens)	13
3.1	Screen-1: Registration Form	13
3.2	Screen-2: Loan Status.....	15
3.3	Screen-3: Update Customer Profiles.....	17
3.4	Screen – 4: Review Loan Applications	19
	Screen – 5: Apply For Loan	21
4	Database design.....	23
4.1	List of Tables	23
5	Stories and Scenario	25
5.1	Story-1: Add New Loan Type in Loan Catalogue.....	25
5.1.1	Scenario# S1.1.....	25
5.1.2	Scenario# S1.2.....	25
5.2	Story-2: Generate Reports	26
5.2.1	Scenario# S2.1.....	26
5.2.2	Scenario# S2.2.....	26
5.3	Story-3: Repay Loan	27
5.3.1	Scenario# S3.1.....	27
5.3.2	Scenario# S3.2.....	27
6	Test Cases	28
7	References	34

List of Figures

Figure 2.1-1 Use case diagram for loan management system	4
Figure 2.2-1 Activity diagram for taking loan	5
Figure 2.3-1 Sequence diagram for taking loan	8
Figure 2.4-1 State diagram for loan.....	9
Figure 2.4-2 State diagram for payment	9
Figure 2.5-1 Class diagram for Library management system	10
Figure 2.6-1 Context diagram for Library management system	11
Figure 2.6-2 DFD level-1 for Library management system.....	12
Figure 2.6-3 DFD level-2 for Issue book	12
Figure 3.1-1 Screen-1: Registration Form	13
Figure 3.2-1 Screen-2: Loan Status.....	16
Figure 3.3-1 Screen-3: Update Customer Profiles.....	17
Figure 3.4-1 Screen-4: Review Loan Applications	19
Figure 3.5-1 Screen-5: Apply For Loan	21

List of Tables

Table 3.1-1 Screen element of Registration form	13
Table 3.2-1 Screen element of Loan Status.....	16
Table 3.3-1 Screen element of Update Customer Profiles.....	17
Table 3.4-1 Screen element of Review Loan Applications	19
Table 3.5-1 Screen element of Apply For Loan	22
Table 4.1-1 Table: Loan Product.....	23
Table 4.1-2 Table: Users	23
Table 4.1-3 Table: Loan Officer	23
Table 4.1-4 Table: Repayment.....	24
Table 4.1-5 Table: Statements.....	24

1 Introduction

1.1 Product perspective

The **Loan Management System (LMS)** is designed to streamline the processes involved in managing loans, from origination to closure. It serves as a comprehensive solution for financial institutions, providing robust functionality to automate and manage loan-related operations efficiently.

1.2 Product features

1.2.1 There are three different users who will be using this product:

- **Admin:** Oversees the entire system and has full control over its functionality.
- **Loan Officer:** Handle loan operations, customer interactions, and compliance.
- **Customer:** Individuals applying for loan and view their loans.

1.2.2 The features that are required for the Admin are:

- Oversee all loan operations and monitor system activities.
- Add, edit, or delete loan products, including their terms (interest rates, tenure, and eligibility criteria).
- Monitor system users (Loan Officers and Customers), including account creation, modification, and deactivation.
- Make and analyse detailed reports, including loan disbursement, overdue accounts, and revenue data.
- Configure system settings, including repayment penalties, grace periods, and automated notifications.
- Monitor compliance with regulatory requirements and generate audit reports.
- Handle system backups and data recovery processes.
- Assign roles and permissions for Loan Officers.
- Monitor the performance of Loan Officers through task and target tracking.

1.2.3 The features that are required for the Loan officer are:

- Review and process loan applications, including document verification and credit assessment.
- Approve or reject loans based on customer eligibility and internal guidelines.
- Track the status of loans, including disbursement and repayments.
- View detailed customer profiles, including their loan history, repayment behaviour, and credit scores.
- Send reminders and notifications to customers regarding due dates and overdue payments.
- Generate customer-specific reports, such as outstanding balances and repayment schedules.
- Respond to customer queries or requests for loan restructuring or early closure.
- Record and track interactions with customers for future reference.
- Monitor pending tasks and customer escalations assigned by the Admin.

1.2.4 The features that are required for the Customer are:

- View available loan products with details such as interest rates, tenure, and eligibility requirements.
- Apply for loans by submitting necessary documentation and filling out the application form.

- Track the status of loan applications and receive updates on approvals or rejections.
- Access their loan account to view repayment schedules, outstanding balances, and due dates.
- Make loan repayments through integrated payment gateways or other channels.
- Request changes to loan terms, such as tenure extension or restructuring.
- Access notifications and reminders for upcoming repayments or overdue amounts.
- Update personal details such as address, contact information, or bank account details.

1.3 Functional Requirement

1.3.1 Admin

- **Add Loan Product:** Add new loan products with details such as interest rate, tenure, eligibility criteria, and terms and conditions to the system.
- **Update Loan Product:** Modify existing loan products if there are changes in terms, rates, or eligibility criteria.
- **Delete Loan Product:** Remove discontinued or incorrect loan products from the system.
- **Store Users:** Maintain a database of all system users, including Loan Officers and Customers, with details like names, IDs, and permissions.
- **Set User Permissions:** Assign specific roles and access levels to Loan Officers.
- **Generate Reports:** Create detailed reports on loans, overdue accounts, revenue, and regulatory compliance.
- **Backup and Recovery:** Take system backups and ensure data recovery in case of failures.
- **Audit Logs:** Track all system activities and modifications for compliance and accountability.
- **Monitor Loan Officer Performance:** Track tasks and performance metrics of Loan Officers.

1.3.2 Loan Officer

- **Authenticate:** Loan Officers must log in with valid credentials to access the system.
- **Review Loan Applications:** Assess customer loan applications, verify submitted documents, and evaluate eligibility criteria.
- **Approve/Reject Loans:** Decide the approval or rejection of loan applications based on customer creditworthiness and institutional guidelines.
- **Manage Customer Profiles:** Access and update customer records, including personal details and loan histories.
- **Track Loans:** View the lifecycle of loans, including application, disbursement, and repayment.
- **Send Notifications:** Notify customers about due dates, overdue payments, or changes in loan terms.
- **Generate Customer Reports:** Create detailed reports specific to individual customers, such as repayment schedules and outstanding balances.
- **Request Loan Restructuring:** Process customer requests for changes in loan terms, such as tenure extension or repayment restructuring.
- **Handle Customer Queries:** Respond to customer inquiries about loans or system-related issues.

1.3.3 Customer

- **Authenticate:** Customers must log in using valid credentials to access their accounts.

- **Apply for Loan:** Submit loan applications with all required information and supporting documents.
- **Track Loan Application:** View the status of their loan application, including updates on approvals or rejections.
- **View Loan Details:** Access detailed information about their loans, including repayment schedules, interest rates, and outstanding balances.
- **Make Repayments:** Pay EMIs or clear overdue payments through integrated payment gateways.
- **Request Loan Changes:** Submit requests for loan restructuring, early closure, or increases in loan amounts.
- **Search Loan Products:** Browse available loan products and calculate EMIs using a loan calculator.
- **Access Notifications:** View alerts for upcoming due dates, payment confirmations, and system updates.
- **Update Personal Details:** Update account information, including address, contact details, or linked bank accounts.
- **Download Statements:** Retrieve loan account statements and payment receipts for record-keeping.

1.4 Non-Functional Requirement

1.4.1 Usability:

- The UI should be simple enough for everyone to understand and get the relevant information without any special training. Different languages can be provided based on the requirements.

1.4.2 Accuracy:

- The data stored about the books and the fines calculated should be correct, consistent, and reliable.

1.4.3 Availability:

- The System should be available for the duration when the library operates and must be recovered within an hour or less if it fails. The system should respond to the requests within two seconds or less.

1.4.4 Maintainability:

- The software should be easily maintainable and adding new features and making changes to the software must be as simple as possible. In addition to this, the software must also be portable.

2 Design and Implementation Constraints

2.1 Use case diagram



Figure 2.1-1 Use case diagram for loan management system

2.2 Activity diagram and Swimlane diagram

SRS – Loan Management System

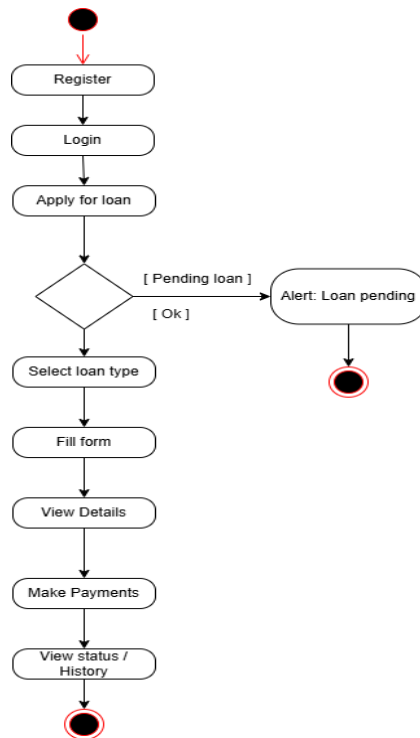


Figure 2.2-1 Activity diagram for taking loan

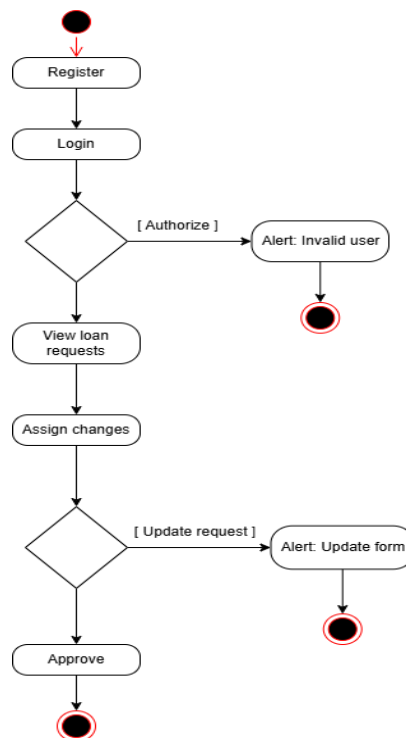


Figure 2.2-2 Activity diagram for approving loan

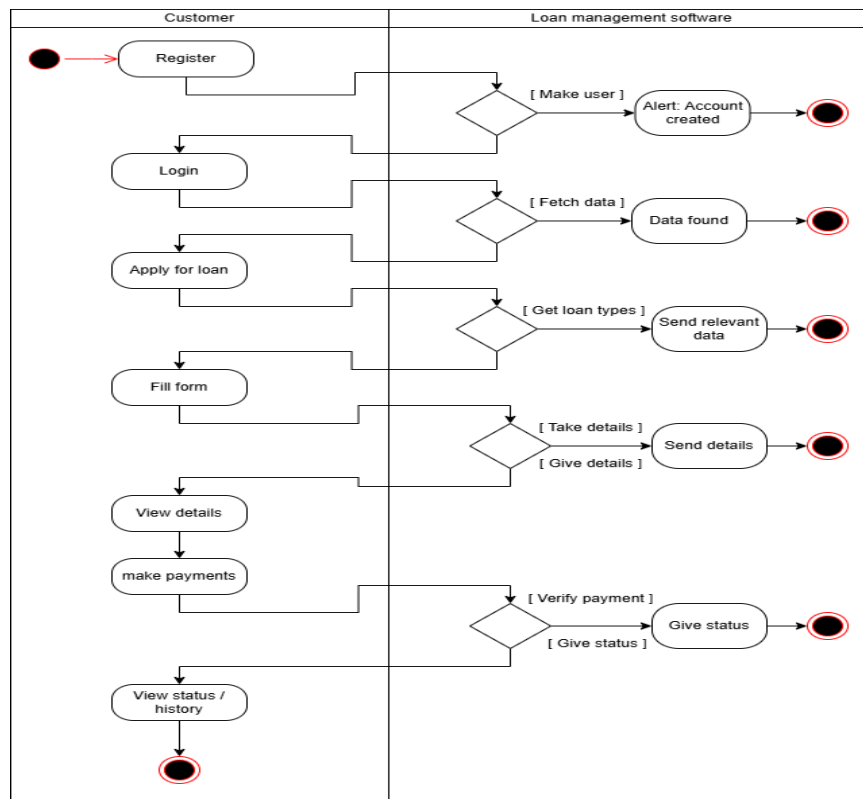


Figure 2.2-3 Swimlane diagram for taking loan

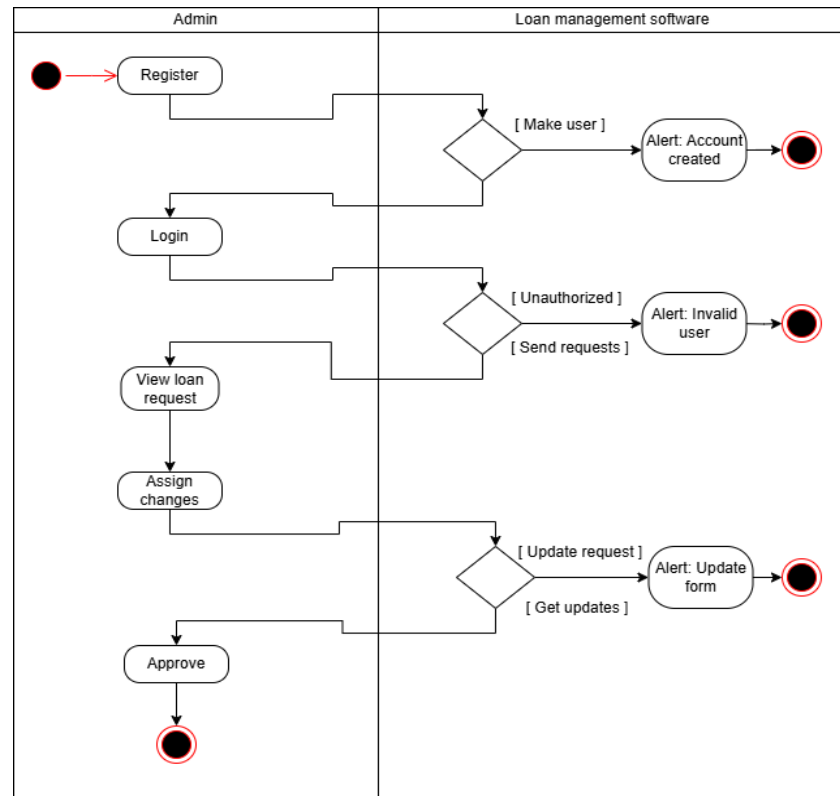


Figure 2.2-4 Swimlane diagram for approving loan

2.3 Sequence diagram

SRS – Loan Management System

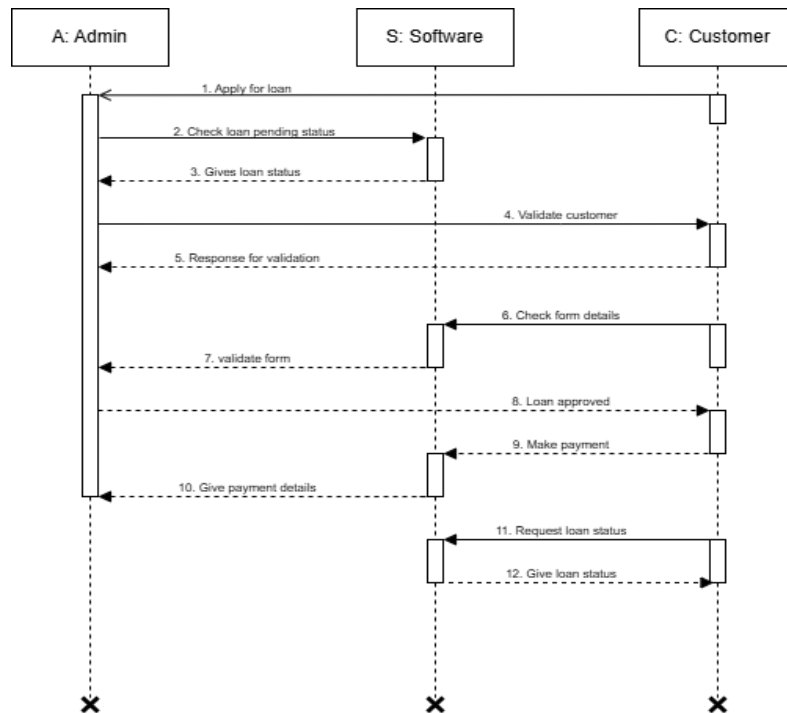


Figure 2.3-1 Sequence diagram for taking loan

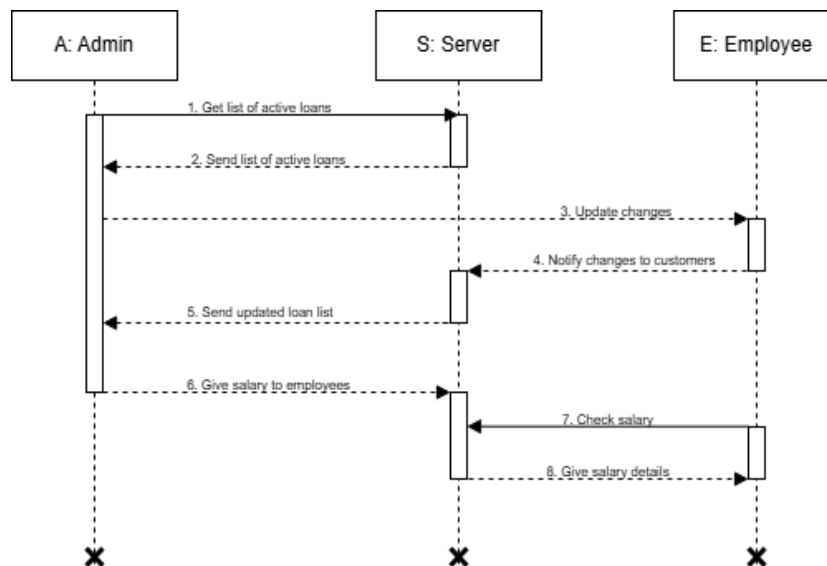


Figure 2.3-2 Sequence diagram for admin user

2.4 State diagram

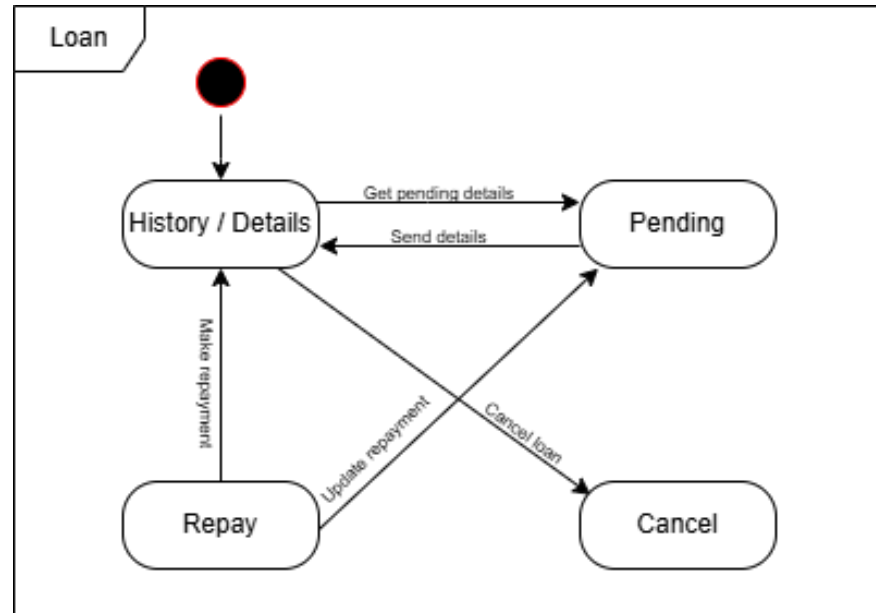


Figure 2.4-1 State diagram for loan

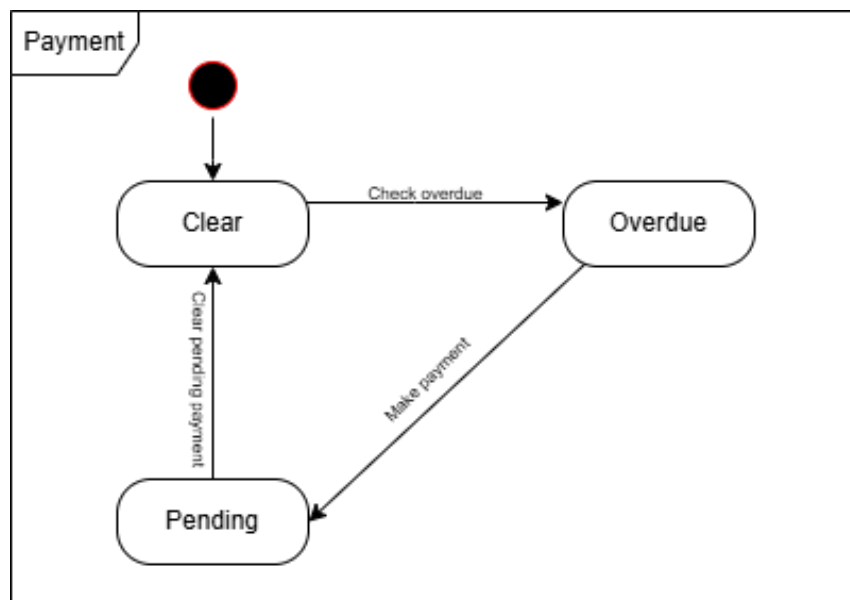


Figure 2.4-2 State diagram for payment

2.5 Class diagram

SRS – Loan Management System

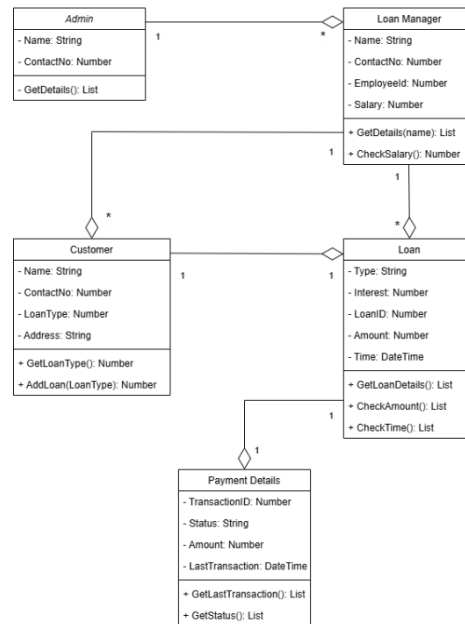


Figure 2.5-1 Class diagram for Library management system

2.6 Data flow diagram

2.6.1 Context diagram (level-0)

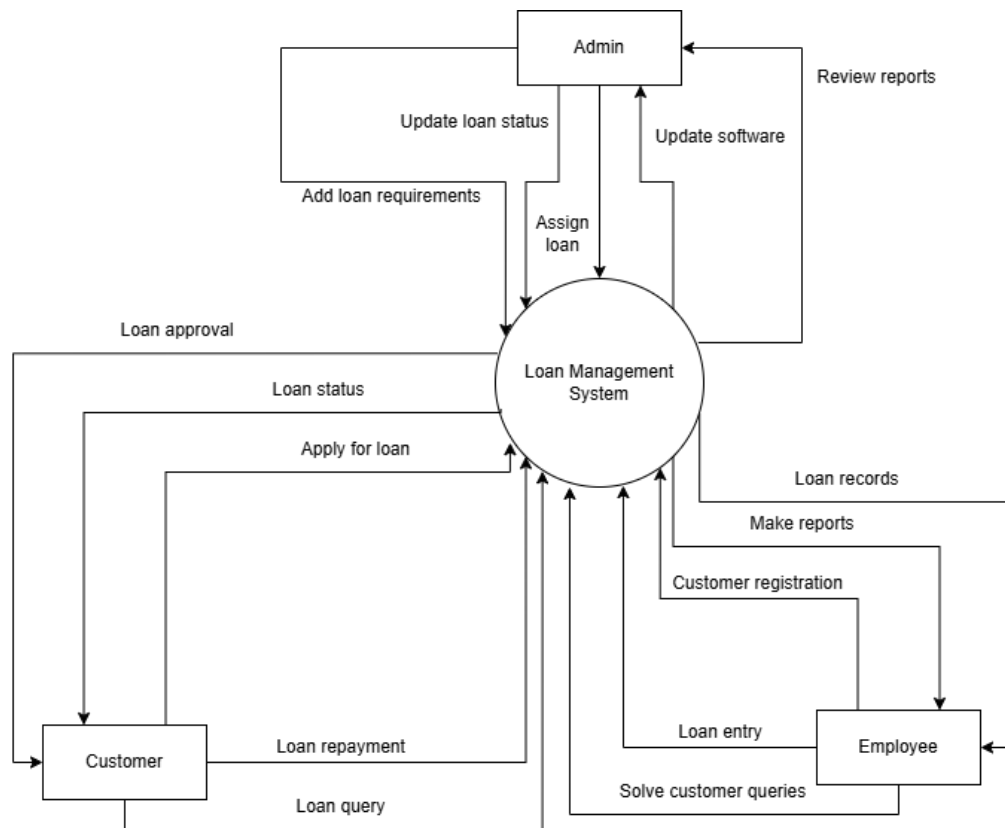


Figure 2.6-1 Context diagram for Library management system

2.6.2 DFD Level-1

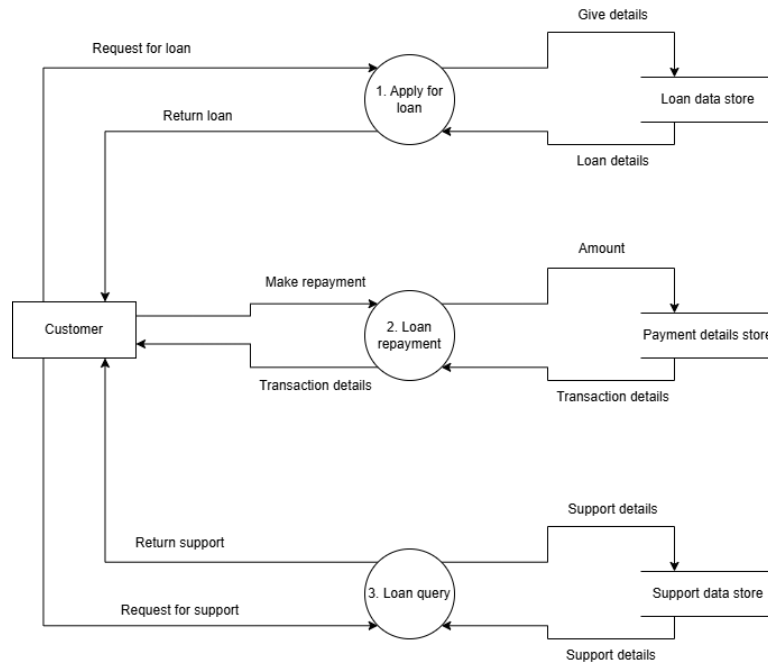


Figure 2.6-2 DFD level-1 for Library management system

2.6.3 DFD Level-2

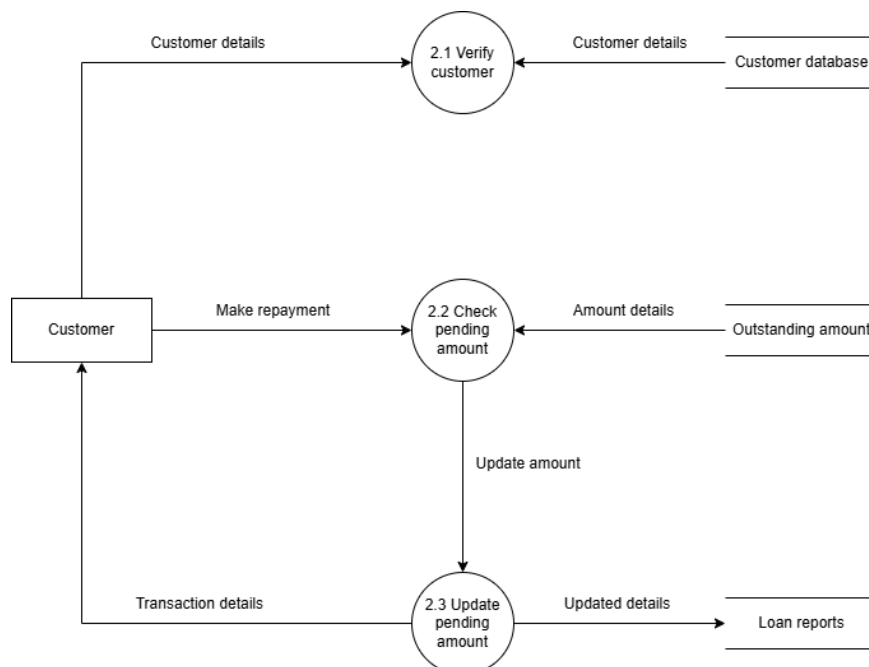
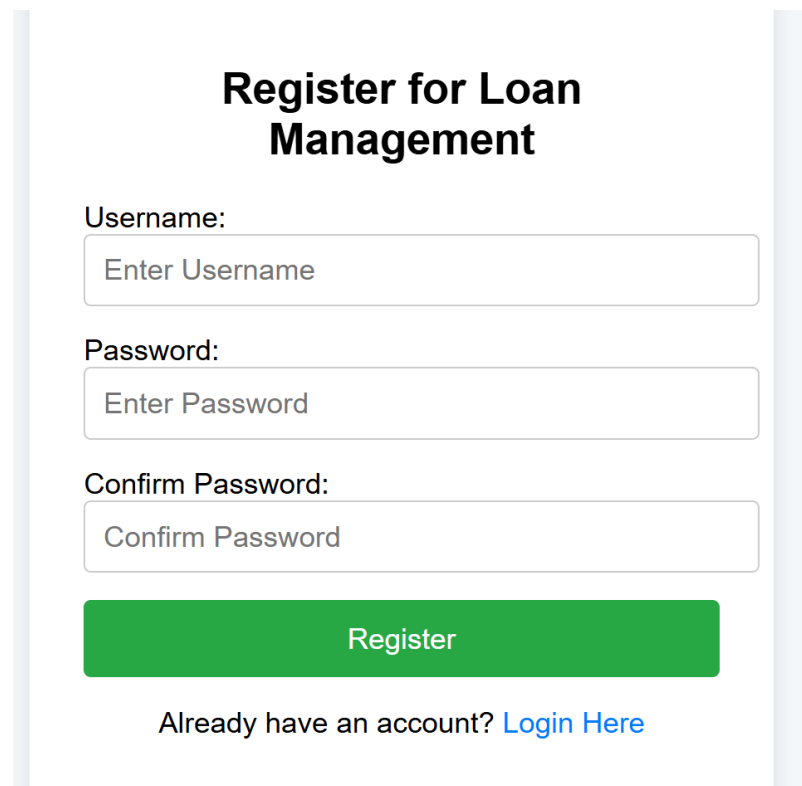


Figure 2.6-3 DFD level-2 for Issue book

3 External interface requirement (Screens)

3.1 Screen-1: Registration Form



The screenshot shows a web form titled "Register for Loan Management". It contains three input fields: "Username:" with a placeholder "Enter Username", "Password:" with a placeholder "Enter Password", and "Confirm Password:" with a placeholder "Confirm Password". Below these fields is a green "Register" button. At the bottom, there is a link that says "Already have an account? [Login Here](#)".

Figure 3.1-1 Screen-1: Registration Form

Purpose: This form will allow the target end-users to register in the system. To register , the following information will be encoded in the system.

Table 3.1-1 Screen element of Registration form

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Username	Textbox	M	1	Username field should be editable and accept the Username.
2	Email	Textbox	M	1	Email field should be editable and accept the email with proper format.
3	Password	Password	M	1	Password field should be editable and accept the password and display as star or dot.

4	Confirm password	Password	M	1	Confirm Password field should be editable and accept the password and display as star or dot.
6	Register	Button	-----	-----	Register is a button for store the entered data into database.
7	Login Here	Button	-----	-----	Login Here is a button to change the screen from register to login

3.2 Screen-2: Loan Status

Your Loan Status

Username	JohnDoe123
Address	123 Main Street, City, Country
Loan Type	Home Loan
Loan Amount	\$50,000
Outstanding Amount	\$30,000
Interest Rate	4.5%
Loan Approval Date	2024-05-10
Time to Repay (months)	36 months
Remaining Repayment Period	18 months
Monthly Repayment	\$833.33
Total Paid	\$20,000
Payment Method	Credit Card
Next Payment Due Date	2025-02-01
Last Payment Amount	\$1,500

Loan Progress:

60% of the loan has been paid off.

Important: Your next payment is due on **2025-02-01**. Please ensure the payment is made on time to avoid penalties.

Make a Payment

Download Statement

Back to Dashboard

Go to Loan History

Figure 3.2-1 Screen-2: Loan Status

Purpose: This screen allows users to view the current status of their loan, including key details such as the loan amount, outstanding balance, monthly repayments, and next payment due date. Users can also track their loan progress via a visual progress bar and receive important notifications regarding their repayment schedule. Additionally, this screen provides options to make a payment, download a loan statement, or navigate back to the dashboard.

Table 3.2-1 Screen element of Loan Status

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Username	Textbox	M	1	Displays the username of the user.
2	Address	Textbox	M	1	Displays the address of the user.
3	Loan Type	Text	M	1	Displays the type of loan.
4	Loan Amount	Text	M	1	Displays the total loan amount granted to the user.
5	Outstanding Amount	Text	M	1	Displays the remaining loan balance that is yet to be paid.
6	Interest Rate	Text	M	1	Displays the interest rate applied to the loan.
7	Loan Approval Date	Text	M	1	Displays the date on which the loan was approved.
8	Time to Repay	Text	M	1	Displays the total repayment time in months.
9	Remaining Repayment Period	Text	M	1	Displays the remaining repayment period in months.
10	Monthly Repayment	Text	M	1	Displays the amount that the user needs to repay monthly.
11	Total Paid	Text	M	1	Displays the total amount the user has already paid toward the loan.
12	Payment Method	Text	M	1	Displays the payment method used by the user.
13	Next Payment Due Date	Date	M	1	Displays the next due date for the loan repayment.
14	Last Payment Amount	Text	M	1	Displays the amount of the last payment made by the user.
15	Loan Progress Bar	Progress Bar	M	1	Displays the loan repayment progress visually.
16	Notifications	Text	O	1	Displays important notifications related to the loan status.
17	Make a Payment	Button	-----	-----	Button to redirect the user to a payment gateway.
18	Download Statement	Button	-----	-----	Button to allow the user to download the loan statement.
19	Back to Dashboard	Button	-----	-----	Button to return the user to the dashboard or previous page.
20	Go to Loan History	Link	-----	-----	Link to navigate to the loan history page.

3.3 Screen-3: Update Customer Profiles

Loan Management Admin [Logout](#)

Search by Customer Name, ID, or Loan Number

Filter by Loan Status Filter by Loan Type Filter by Customer Status

Customer ID	Name	Phone	Email	Loan Type	Loan Amount	Loan Status	Last Loan Date	Actions
12345	John Doe	+1 234 567 890	johndoe@example.com	Personal	\$5000	Active	2025-01-10	Edit View

Customer Profile: John Doe

Name:

Phone:

Email:

Loan Type:

Loan Amount:

Loan Status:

Loan History

Loan Type	Loan Amount	Status	Start Date
Personal	\$5000	Active	2025-01-10

Figure 3.3-1 Screen-3: Update Customer Profiles

Purpose: This screen allows admins to search, filter, and view customer loan information. Admins can edit customer profiles and access loan histories for efficient oversight.

Table 3.3-1 Screen element of Update Customer Profiles

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Customer ID	Textbox	M	1	Displays the unique customer ID for reference.
2	Customer Name	Textbox	M	1	Displays and allows editing of the customer's name.
3	Customer Phone	Textbox	M	1	Displays and allows editing of the customer's phone number.
4	Customer Email	Textbox	M	1	Displays and allows editing of the customer's email address.
5	Loan Type	Textbox	M	1	Displays the loan type (Personal, Auto, Home) for the customer.
6	Loan Amount	Number Input	M	1	Displays and allows editing of the loan amount for the customer.
7	Loan Status	Textbox	M	1	Displays the current loan status (Active, Closed, Pending).

8	Last Loan Date	Date Input	M	1	Displays the last loan date (when the loan was granted).
9	Edit Button	Button	----	1	Button to edit the customer's loan information.
10	View Button	Button	----	1	Button to view the detailed loan and customer information.
11	Customer Profile - Name	Textbox	M	1	Editable text field for the customer's name in the profile section.
12	Customer Profile - Phone	Textbox	M	1	Editable text field for the customer's phone number in the profile section.
13	Customer Profile - Email	Textbox	M	1	Editable text field for the customer's email address in the profile section.
14	Loan Type (Profile Section)	Textbox	M	1	Editable text field for the loan type (e.g., Personal, Auto) in the customer profile section.
15	Loan Amount (Profile Section)	Textbox	M	1	Editable text field for the loan amount in the customer profile section.
16	Loan Status (Profile Section)	Textbox	M	1	Editable text field for the loan status (e.g., Active, Pending) in the customer profile section.
17	Save Changes Button	Button	----	1	Button to save the edited customer details and loan information.
18	Loan History Table	Table	M	1	Displays a table with loan history information for the customer, including loan type, amount, and status.
19	Loan History - Loan Type	Textbox (Read-Only)	----	N	Displays the type of loan for each entry in the loan history.
20	Loan History - Loan Amount	Textbox (Read-Only)	----	N	Displays the amount of the loan in the loan history table.
21	Loan History - Loan Status	Textbox (Read-Only)	----	N	Displays the loan status for each loan entry in the loan history table.
22	Loan History - Start Date	Date Input (Read-Only)	----	N	Displays the start date of each loan in the loan history table.

3.4 Screen – 4: Review Loan Applications

Review Loan Applications

Search by applicant name.. Filter by Status ▾ Search

Applicant Name	Loan Amount	Loan Type	Status	Date Applied	Actions
John Doe	5000	Personal Loan	Pending	2025-01-15	View Approve Reject
Jane Smith	15000	Home Loan	Approved	2025-01-10	View
Mark Lee	7000	Auto Loan	Rejected	2025-01-12	View
Emily Davis	20000	Home Loan	Pending	2025-01-14	View Approve Reject
Michael Brown	8000	Personal Loan	Approved	2025-01-13	View

[Previous](#) [1](#) [2](#) [3](#) [Next](#)

Figure 3.4-1 Screen-4: Review Loan Applications

Purpose: This screen allows users to review loan applications, search and filter by applicant name or loan status, and take action on pending applications. Admins can approve or reject loans, view application details, and navigate through paginated results to efficiently manage loan review tasks.

Table 3.4-1 Screen element of Review Loan Applications

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Search by Applicant Name	Textbox	O	1	Input field for searching by applicant name.
2	Filter by Status	Dropdown (Select)	O	1	Dropdown to filter loan applications by status (Pending, Approved, Rejected).
3	Search Button	Button	----	1	Button to trigger the search functionality based on the entered name and selected status.
4	Applicant Name (Table Column)	Static Text	----	N	Displays the name of the loan applicant in the table.
5	Loan Amount (Table Column)	Static Text	----	N	Displays the loan amount in the table.
6	Loan Type (Table Column)	Static Text	----	N	Displays the type of loan in the table.
7	Loan Status (Table Column)	Static Text (Status Badge)	----	N	Displays the loan status with a badge (Pending, Approved, Rejected).

8	Date Applied (Table Column)	Static Text	----	N	Displays the date when the loan was applied.
9	View Loan Button	Button	----	N	Button to view the details of a specific loan application.
10	Approve Loan Button	Button	O	N	Button to approve a loan application (only visible for pending applications).
11	Reject Loan Button	Button	O	N	Button to reject a loan application (only visible for pending applications).
12	Previous Pagination Button	Button/Link	----	1	Button/link to navigate to the previous page of loan applications.
13	Page Number (Pagination)	Button/Link	----	N	Buttons for pagination to move between pages of loan applications.
14	Next Pagination Button	Button/Link	----	1	Button/link to navigate to the next page of loan applications.

Screen – 5: Apply For Loan

Apply for a Loan

Username:

Address:

Loan Type:



Loan Amount:

Repayment End Date:



Payment Method:

☐ Credit Card ☐ Debit Card

CVV (for card payments):

Figure 3.4-2 Screen-5: Apply For Loan

Purpose: This page allows users to apply for a loan by providing personal details, selecting a loan type, specifying the loan amount, and choosing a repayment date. Users can also select a payment method and provide CVV details for card payments. Upon submission, the application calculates the loan repayment period and alerts the user.

Table 3.4-2 Screen element of Apply For Loan

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Username	Textbox	M	1	Username field should be editable and accept the Username.
2	Address	Textbox	M	1	Address field should be editable and accept the Address.
3	Loan Type	Dropdown	M	1	Loan Type field should be editable and allow the user to select from options.
4	Loan Amount	Number Input	M	1	Loan Amount field should be editable and accept numeric values for the loan amount.
5	Repayment End Date	Checkbox Group	M	1	Repayment End Date field should be editable and allow the user to select a date.
6	Payment Method	Checkbox Group	O	1	A group of checkboxes to select payment methods.
7	CVV	Password	M	1	CVV field should be editable and accept the 3 or 4 digit CVV.
8	Submit	Button	-----	-----	Submit button to send the loan application data to the backend.

4 Database design

4.1 List of Tables

- Loan Product
- Users
- Loan Officer
- Repayment
- Statements

Table 4.1-1 Table: Loan Product

Column	Data Type	Null	Keys & Constrains	Default Value & Description
LoanID	Int	NN	PK (Auto Increment)	
LoanName	varchar(100)	NN		
InterestRate	Decimal(8,2)	NN		
EligibilityCriteria	Varchar(100)	NN		
OfficerID	Int	NN	FK	Reference to Loan Officer

Table 4.1-2 Table: Users

Column	Data Type	Null	Keys & Constrains	Default Value & Description
UserID	Int	NN	PK (Auto Increment)	
UserName	varchar(100)	NN		
Contact	Int	NN		
Address	Varchar(100)	NN		
PendingLoan	Int	AN	FK	Reference to Loan Product
Permissions	Varchar(100)	AN		

Table 4.1-3 Table: Loan Officer

Column	Data Type	Null	Keys & Constrains	Default Value & Description
EmpID	Int	NN	PK (Auto Increment)	
EmpName	varchar(100)	NN		
DepartmentID	Int	NN	FK	Reference to Loan Product
DepartmentName	Varchar(100)	NN		
Salary	Int	NN		

Table 4.1-4 Table: Repayment

Column	Data Type	Null	Keys & Constrains	Default Value & Description
TransactionID	Int	NN	PK (Auto Increment)	
UserID	Int	NN	FK	Reference to Users
LoanID	Int	NN	FK	Reference to Loan Product
Overdue	Int	NN		

Table 4.1-5 Table: Statements

Column	Data Type	Null	Keys & Constrains	Default Value & Description
StatementID	Int	NN	PK (Auto Increment)	
UserID	Int	NN	FK	Reference to Users
TranclD	Int	NN	FK	Reference to Repayment
StatementDetails	Varchar(100)	NN		

5 Stories and Scenario

5.1 Story-1: Add New Loan Type in Loan Catalogue

Story # S1	: As a Admin, I want to add a new loan type in loan catalogue. So that everyone can easily find and apply for it.
Priority	: High
Estimate	: XL
Reason	: The addition of a new loan type to the loan catalogue is crucial for ensuring that the loan's collection is up-to-date and accessible to everyone.

5.1.1 Scenario# S1.1

Scenario# S1.1	: Adding a New Loan Type with Valid Information
Prerequisite	: Admin is logged in to the Loan management system.
Acceptance Criteria	Given: The Admin is navigated to the loan catalogue page. Valid loan information, including title, interest rates, tenure, and other relevant details is added. When: The admin selects the "Add New Loan Type" option And The admin enters valid Loan details The admin clicks the "Save" button to add the loan type to the catalogue. Then the system successfully adds the loan type to the catalogue and the admin receives a confirmation message with the loan's identification number.

5.1.2 Scenario# S1.2

Scenario# S1.2	: Adding a New Loan Type with Invalid Information.
Prerequisite	: The admin is logged into the loan management system.

Acceptance Criteria	Given: The admin is on the loan catalogue management page When: The admin selects the "Add New Loan Type" option and the admin enters an incomplete or incorrect loan details and admin clicks the "Save" button to add the loan type to the catalogue. Then the system displays error messages for the incorrect or missing information and the loan type is not added to the catalogue.
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5.2 Story-2: Generate Reports

Story # S2	As a Loan officer, I want to generate reports for active loans of users. So that I can review the loans of users and suggest changes.
Priority	High
Estimate	M
Reason	Generating reports is crucial for monitoring loan performance and assisting users with better financial planning.

5.2.1 Scenario# S2.1

Scenario# S2.1	Generate Active Loan Report
Prerequisite	Loan officer is logged into the system.
Acceptance Criteria	Given: The loan officer navigates to the "Reports" section. When: The officer selects the "Active Loans Report" option and specifies a date range. Then The system generates a report displaying all active loans with details such as customer name, loan amount, and due dates.

5.2.2 Scenario# S2.2

Scenario# S2.2	Filter Reports by Loan Type
Prerequisite	Loan officer is logged into the system.
Acceptance Criteria	Given: The loan officer accesses the "Reports" section. When: The officer selects "Filter by Loan Type" and chooses a loan category.

	Then: The system generates a report displaying only loans of the selected type, allowing detailed analysis.
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5.3 Story-3: Repay Loan

Story # S3	:	As a customer, I want to repay my loan through the system, So that I can clear my dues conveniently and avoid penalties.
Priority	:	High
Estimate	:	M
Reason	:	A streamlined repayment process enhances customer experience and ensures timely loan settlements.

5.3.1 Scenario# S3.1

Scenario# S3.1	:	Successful Loan Repayment
Prerequisite	:	Customer is logged into the Loan Management System and has an active loan.
Acceptance Criteria	:	Given: The customer navigates to the "Repay Loan" section. When: The customer selects a loan, enters the repayment amount, and confirms the transaction. Then The system processes the payment, updates the outstanding loan balance, and sends a confirmation message.

5.3.2 Scenario# S3.2

Scenario# S3.2	:	Attempting Repayment with Insufficient Funds
Prerequisite	:	Customer is logged into the system and attempts to make a loan repayment.
Acceptance Criteria	:	Given: The customer selects a loan and enters the repayment amount. When: The system detects that the account has insufficient funds. Then: The system displays an error message and prevents the transaction from being processed.

6 Test Cases

Project Name:	LMS	Test Designed by:	Kalp V. Viradia
Module Name:	Login	Test Designed date:	19-02-2025
Release Version:	1.0	Test Executed by:	R. B. Gondaliya
		Test Execution date:	19-02-2025

Pre-condition: Web application should be accessible				
Test Case ID	Test Title	Test Type	Description	Test Case ID
TC_001	Login to Loan Management System with valid credentials	Functional	Login to Loan Management System web application through valid credentials	TC_001
TC_002	Login to web application with invalid credential	Functional	Login to Loan Management System web application through invalid credentials	TC_002

Test Case Title	Login to Loan Management System with valid credentials
Test Type	Functional
Test Priority	High
Pre-condition	Web application should be accessible. The user must have valid credentials.

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Access Web application URL	The site launched properly	Site launched successfully	Pass		https://accounts.google.com/ServiceLogin	
2	Enter valid Username in username field	Username field should be editable and accept the Username	Username input accepted	Pass		Username: Rbgondaliya@gmail.com	
3	Enter valid Password in Password field	Password field should be editable and accept the password and display as star or dot	Password input displayed in dots and accepted	pass		Password: rbgondaliya	
4	Enter valid captcha code in captcha field	Captcha field should be editable and accept captcha and	Captcha input accepted	Pass	Step required when human action	get captcha from image which is near by captcha field	

		captcha is case sensitive			validation perform		
5	Click on login button	User should login into site and navigated to dashboard	User navigated to dashboard and username displayed on top of the right side.	pass			

Test Case Title	Login to web application with invalid credentials
Test Type	Functional
Test Priority	Medium
Pre-condition	Web application should be accessible.

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Verify login with invalid username and invalid password	An error message should appear indicating wrong username or password	Error message displayed as "Wrong username or password"	Pass		Invalid Username: testuser@xyz.com, Invalid Password: testpassword	
2	Verify login with valid username and invalid password	An error message should appear indicating incorrect password	Error message displayed as "Incorrect password"	Pass		Username: rbgondaliya@gmail.com, Invalid Password: wrongpassword	
3	Verify login with invalid username and valid password	An error message should appear indicating "Username not found"	Error message displayed as "Username not found"	Pass		Invalid Username: testuser@xyz.com, Valid Password: rbgondaliya	

Project Name:	LMS	Test Designed by:	Kalp V. Viradia
Module Name:	Loan Repayment Process	Test Designed date:	19-02-2025
Release Version:	1.0	Test Executed by:	R. B. Gondaliya
		Test Execution date:	19-02-2025

Pre-condition: Web application should be accessible

Test Case ID	Test Title	Test Type	Description	Test Case ID
TC_001	Loan Repayment Process	Functional	Test the loan repayment functionality to ensure the user can repay the loan successfully	TC_001
TC_002	Loan Repayment with Invalid Amount	Functional	Ensure that the system validates repayment amounts that are invalid	TC_002

Test Case Title	Loan Repayment Process
Test Type	Functional
Test Priority	High
Pre-condition	Test the loan repayment functionality to ensure the user can repay the loan successfully

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Navigate to the Loan Repayment section from the dashboard	Loan Repayment section should open successfully	Loan Repayment section opened successfully	Pass			
2	View outstanding loan balance	The outstanding loan balance should be displayed	Outstanding balance of \$5000 shown	Pass			
3	Enter repayment amount in the payment field	The user should be able to enter the repayment amount	Repayment amount of \$1000 entered successfully	Pass	Payment field is editable	Amount: \$1000	
4	Select payment method (e.g., Credit Card, Bank Transfer)	The user should be able to select a payment method from the dropdown	Payment method selected as Credit Card	Pass	Payment method dropdown function	Credit Card	

					ing correctl y		
5	Click on the "Repay Loan" button	Loan repayment should be processed successfully, and a confirmation message should appear	Loan repayment of \$1000 processed successfully with confirmation message	Pass	Confirm ation message : "Your loan repaym ent of \$1000 has been successf ully process ed."		
6	Verify the updated loan balance	The loan balance should reflect the repayment amount and update accordingly	New balance displayed as \$4000	Pass	Balance updated successf ully after repaym ent		
7	Verify email/SMS notification	User should receive a confirmation email or SMS regarding the loan repayment	Confirmation email received	Pass	Email received at rbgonda liya@g mail.co m		

Test Case Title	Login to web application with invalid credentials
Test Type	Functional
Test Priority	Medium
Pre-condition	Web application should be accessible. The user must be logged in and have an active loan with outstanding payments.

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comme nt	Data	BUG ID
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1	Navigate to the Loan Repayment section from the dashboard	Loan Repayment section should open successfully	Loan Repayment section opened successfully	Pass			
2	View outstanding loan balance	The outstanding loan balance should be displayed	Outstanding balance of \$5000 shown	Pass			
3	Enter an invalid repayment amount (e.g., negative or zero)	The system should reject the invalid amount and display an error message	Error message displayed: "Repayment amount must be a positive value"	Pass	Validation working as expected	Amount: -\$100 or \$0	
4	Attempt to proceed with invalid repayment amount	The system should not allow proceeding with the invalid repayment	Button to proceed remains disabled or hidden	Pass			
5	Enter a valid repayment amount (e.g., \$1000)	The repayment amount should be accepted	Valid repayment amount of \$1000 entered	Pass		Amount: \$1000	
6	Proceed with the loan repayment	The loan repayment should be processed successfully	Loan repayment processed and confirmation displayed	Pass	Confirmation message : "Your loan repayment of \$1000 has been successfully processed."		
7	Verify updated loan balance	The loan balance should reflect the repayment amount and update accordingly	New balance displayed as \$4000	Pass			

8	Verify email/SMS notification	User should receive a confirmation email or SMS regarding the loan repayment	Confirmation email received	Pass	Email received at rbgondaliya@gmail.com		
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7 References

- http://www.w3schools.com/html/html_intro.asp
- <https://www.w3schools.com/php/default.asp>
- <https://www.javatpoint.com/uml>

SE Lab

GET Get

```
https://api.openweathermap.org/data/3.0/onecall?lat={33.44}&lon={-94.04}&exclude={part}&appid={705879fbf4706cf7431f7ade00682851}
```

StartFragment

To get access to current weather, minute forecast for 1 hour, hourly forecast for 48 hours, daily forecast for 8 days and government weather alerts, please use this section of the documentation.

If you are interested in other functionality on One Call API 3.0, please check [Product concept](#) to follow the right section.

EndFragment

PARAMS

lat	{33.44}
	Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
lon	{-94.04}
	Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
exclude	{part}
	By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces). Available values: current minutely hourly daily alerts
appid	{705879fbf4706cf7431f7ade00682851}
	Your unique API key (you can always find it on your account page under the "API key" tab)

POST Post

`https://api.openweathermap.org/data/3.0/onecall?lat={33.44}&lon={-94.04}&exclude={part}&appid={705879fbf4706cf7431f7ade00682851}`

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EndFragment

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lat	{33.44} Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
lon	{-94.04} Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
exclude	{part} By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces). Available values: current minutely hourly daily alerts
appid	{705879fbf4706cf7431f7ade00682851} Your unique API key (you can always find it on your account page under the "API key" tab)

PUT Put

`https://api.openweathermap.org/data/3.0/onecall?lat={33.44}&lon={-94.04}&exclude={part}&appid={705879fbf4706cf7431f7ade00682851}`

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lon	{-94.04}	Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
exclude	{part}	By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces). Available values: current minutely hourly daily alerts
appid	{705879fbf4706cf7431f7ade00682851}	Your unique API key (you can always find it on your account page under the "API key" tab)

PATCH Patch

`https://api.openweathermap.org/data/3.0/onecall?lat={33.44}&lon={-94.04}&exclude={part}&appid={705879fbf4706cf7431f7ade00682851}`

StartFragment

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EndFragment

PARAMS

lat	{33.44}	Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
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lon	<code>{-94.04}</code>	Longitude, decimal (-180; 180). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
exclude	<code>{part}</code>	By using this parameter you can exclude some parts of the weather data from the API response. It should be a comma-delimited list (without spaces). Available values: current minutely hourly daily alerts
appid	<code>{705879fbf4706cf7431f7ade00682851}</code>	Your unique API key (you can always find it on your account page under the "API key" tab)

DELETE Delete

`https://api.openweathermap.org/data/3.0/onecall?lat={33.44}&lon={-94.04}&exclude={part}&appid={705879fbf4706cf7431f7ade00682851}`

StartFragment

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EndFragment

PARAMS

lat	<code>{33.44}</code>	Latitude, decimal (-90; 90). If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
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current minutely hourly daily alerts

appid

{705879fbf4706cf7431f7ade00682851}

Your unique API key (you can always find it on your account page under the "API key" tab)